Everything CLS does, plus anything that needs a vein, a scalpel or a rhythm decision.

M MASSIVE HEMORRHAGE

- **TXA 1 g IV** for severe bleeding, noticeable bruising or a bleeding hemothorax.
- Bruising with a falling blood pressure is internal bleeding. TXA and fluids.
- Clots are unstable. Wrap or stitch once stable.

A AIRWAY

TOOL	SOLVES
i-gel	Open airway, collapse. Airway must be clear first
ACCUVAC, suction bag	All obstructions
Surgical airway (CricKit)	Collapse, obstruction, inflammation, spasm

Use the least invasive airway that solves the problem.

R RESPIRATION

- **Stethoscope:** muffled on one side is a pneumothorax.
- **Tension pneumothorax:** chest seal plus NCD, or thoracostomy.
- **Hemothorax:** TXA, then thoracostomy and chest tube.
- Consider lidocaine IM before thoracostomy.

FINGER THORACOSTOMY	DO
Lung inflating normally	Close incision
Lung not inflating or collapsed	Close incision
Blood in pleural space	Chest tube
Active bleeding	TXA, then chest tube

Chest tube: drain with the ACCUVAC or a suction bag, re-drain until no blood comes out, close the incision.

H HEAD, PAIN, FRACTURES

- **Unconscious, stable vitals:** ammonia inhalant.
- **Still unconscious:** go back through M, A, R, C.
- **Fracture:** lidocaine 50 to 100 mg IM, realign, SAM splint, check mobility, wrap.
- **Complex fracture:** evacuate.

C ACCESS

ACCESS	SITE	FLOW
16g IV	Limbs	Moderate
14g IV	Limbs	Fast. More complications
FAST1 IO	Sternum	Slow
EZ-IO	Tibia or humerus	Slow

Crashing casualty: IO first, then move the bags to an IV. Tourniquets block IVs on that limb.

C FLUIDS

FLUID, BEST FIRST	CARRIES OXYGEN	FLOW	PLATELETS
Fresh whole blood	Yes	Fast	Yes, with calcium
Whole blood	Yes	Slow	Yes, with calcium
Plasma	30%	Moderate	Yes
Saline	No	Fast	No

BLOOD

- Check the blood type on the dog tag. A mismatch can kill.
- O- goes to anyone. AB+ receives from anyone. A negative recipient only takes negative blood.
- **Calcium chloride 1 g IV** after the first unit (500 ml), then after every 4 units. Never IM.

FIELD TRANSFUSION

1. Connect the kit to the donor's IV. It does not work on an IO.
2. Wait for it to fill.
3. Disconnect and use it as fresh whole blood, soon.

CASUALTY CONVERSION

- Casualty unconscious, alive and not bleeding externally.
- If the mission requires it: in a medical facility or vehicle, and ertapenem 1 g given.
- Keep treating the AI casualty until it is evacuated.

C CARDIAC ARREST

Confirm with a carotid pulse check. Fix the reversible cause. Then rhythm, drug, shock, CPR.

REVERSIBLE CAUSE	DO
Hypovolemia	IO, push fluid, CPR
Hypoxia	CPR with BVM
Tension pneumothorax	NCD or thoracostomy
Hemothorax	TXA, thoracostomy, chest tube
Opioids on board	Naloxone

RHYTHM	SHOCK	DO
Sinus	No	Re-assess
VT or VF	Yes	Amiodarone or lidocaine, shock, CPR
PEA	No	Re-check reversible causes, CPR
Asystole	No	Check for terminal injuries, epinephrine, CPR

PEA looks like sinus on the EKG. Always confirm with a pulse check.

ARREST DRUGS

- **Epinephrine 1 mg IV.** The only drug that helps in asystole.
- **Amiodarone 300 mg IV** for VT or VF. Two vials. Less does nothing.
- **Lidocaine 100 mg IV** for VT or VF. One vial.
- Never amiodarone and lidocaine together. Leave 5 to 10 minutes between them, and between amiodarone doses.

CPR

- 2 minutes, or until the AED prompts. Re-analyze. Repeat.
- **Short of oxygen, one medic or no i-gel:** 30 compressions, then 2 BVM breaths.
- **Short of oxygen, two medics and an i-gel:** continuous CPR and BVM.
- **Pulse is back:** EtCO2 jumps to 40 to 50, cyanosis fades, carotid pulse returns.